University of Florida, Department of Mathematics

Algebra First-Year Exam, January 2016, Part 2

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Let F be a field, let $n \geq 2$, and let R be the ring of $n \times n$ matrices over F.

(a) Show that the only 2-sided ideals of R are $\{0\}$ and R.

(b) Show that R contains a nonzero proper left ideal I.

2. Give an example of each of the following, with some explanation.

(a) A UFD which is not a PID.

(b) A ring R with unit and a ring S with unit such that S is a subring of R but $1_R \neq 1_S$.

(c) A commutative ring R with unit and an irreducible element $\pi \in R$ which is not prime.

3. Prove that the ideal (X, Y) in the ring $\mathbb{Q}[X, Y]$ is not a principal ideal.

4. Let R be a ring and let M be an R-module. Define

$$\mathrm{Tor}(M) = \{x \in M : rx = 0 \text{ for some } r \in R \setminus \{0\}\}.$$

(a) Prove that if R is an integral domain then $\mathrm{Tor}(M)$ is a submodule of M.

(b) Give an example of a ring R and an R-module M such that $\mathrm{Tor}(M)$ is not a submodule of M. Prove any claims that you make.

5. Find a representative for each conjugacy class in $\mathrm{GL}_2(\mathbb{Z}/3\mathbb{Z})$ consisting of elements whose order divides 8. (Note that we have $X^4 + 1 = (X^2 - X - 1)(X^2 + X - 1)$ in $(\mathbb{Z}/3\mathbb{Z})[X]$.)

6. Let L/K be an algebraic field extension and let R be a subring of L that contains K. Prove that R is a field.